Clickboat: Boat repair and rental Management System

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RATIONALE

The nature of boat repair and rental counting are time consuming and requires workforce and more effort to do. Manual process may also be hard to be done on a regular basis. By using the proposed project, boat repair and rental transaction processes will be automated and will be easy to manage on a daily and regular basis. The yacht owner, yacht charter, tourist and boat owners will be able to save time and effort such as notifications and updating transaction status.

OBJECTIVES

The general objective of the study is to develop a web/mobile application that can be used by yacht owner, yacht charter, tourist and boat owners. To be able to understand the main objective, the specific objectives were drawn: (1) Define problems encountered in: a. Yacht repair scheduling ,b.Yacht repair management, c. tourist iterinary ,d. boat scheduling;(2) Define modules of the system developed to address the problem;(3) Define the tools and technology used to develop the solution; (4) the level of acceptance of the respondents in the developed system in terms of: a. user friendliness , functionality ; (5) Issues encountered during the initial deployment of the system;(6) Features recommended to improve the system.

METHODOLOGY

To be able to develop the final output, the study followed the processes involved in the Agile Development. It includes all the processes that is necessary to finish the study which are: (1)Plan , (2)Design, (3)Develop (4)Test, (5) Release , (6) Feedback. The data needed to conduct the study were analysed and validated based from the system requirements of the project.

RESULTS

The system's evaluation resulted with the mean range of 4 above with the equivalent of high acceptability rating for both user friendliness and functionality. The developed modules are for tourist, yacht charter, yacht owner and boat owner where they can communicate and automate some of the process that involves notification and report generation.

CONCLUSIONS

Mobile and web application is a good way to cater marina based services such as boat repair and rental. Different available API and libraries are very useful in the development because it speeds up the process. The ability to test different environment of the system is a must especially technology is evolving.

KEYWORDS:*Yacht charter, Tourist Itinerary, Web application, Boat Scheduling, Yacht repair scheduling*.

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1. INTRODUCTION

The boating and yacht industry has been one of the major components in terms of marina tourism, Boat rental and repair has been vital yacht charter maybe not a typical type of business even though owning yacht is pretty expensive clients are still availing and finding services for boat repairs, same as the boat rental industry which may comprises tourism such as boat ride and island hopping , in addition to that Philippines is an archipelago that makes it a good spot for marina services such as marina tourism where the Tourist can book boat ride on Boat owners for tourists spot islands such as Fortune Island and Twin Island on Nasugbu,Batangas. The Country is also a hub for numerous number of Yacht charter that offers repairs on either boat or ships. Even though in some countries like US there are unemployment rate increase that makes the market on boating to decline but still the industry revenue manages to come back on the market with some constraints on the line such as smaller market but few competitors around due to unemployment of the crews.

Papaya yacht charter is one of the local yacht charter that offers this kind of service on the municipality of Nasugbu, they often use paper and flat database like excel on recording transaction, because of this report generating can take time to do and another is the handling of transactions is hard when it is open season or occasion of yacht repairs and maintenance transactions record are hard to keep because it is on the form of paper records that is probable of it to lost and will not be able to retrieve it. Another is the Boat owners have difficulty in terms of managing the transactions in terms of different tourist, tourist may also encounter difficulties of looking boat owners who has the right size of boat to rent. Barangay Papaya has been a tourist spot on the past years since 2015, manual transaction has been the constraint they have been facing that makes it difficult for the boat owners to cater guests.

The purpose of this research titled “Click Boat : Boat Repair and Rental Management System” is to improve and the business process on the industry of Boat repair and rental with the use of technologies and develop a system to serve both sides of yacht owners,manager, boat owner and tourists to help improve the aspects of the business such as report generating, tracking and management of the information needed to became a functional and efficient service provider. .

2. OBJECTIVES

The research aim to develop ClickBoat: Boat Repair and Rental Management System.
Specifically it sought the following:

1. What are the common problems encountered by the respondents in terms of:
   - 1.1 Tourist’s Itinerary
   - 1.2 Boat Scheduling
   - 1.3 Yacht Repair Scheduling
   - 1.4 Yacht Repair Management
2. What are the modules of the system to be develop to address the common problems encountered by the respondents?
3. What are the tools and technology did the Researchers use to develop the solution to the problem?
4. What is the level of acceptance of the respondents in the developed system “Click Boat: Boat Repair and Rental Management System” in terms of?
   - 4.1 User Friendliness; and
   - 4.2 Functionality
5. What Issues do the researchers encounter during the initial deployment of the “Click Boat: Boat Repair and Rental Management System”?
6. What are the features recommended by the respondents to improve the functionality of the system?

3. LITERATURES REVIEW

In different cases business automation is needed on a business as it grows from medium to large, Yacht repair servicing businesses and boat rentals start to use technologies to improve their performance and bring the customer to a better experience on giving the service, West Marine states that it uses its e-commerce website to test markets for new products in a cost-effective manner. [1]. the web became a famous platform over the past years; benefits that it offers are mutually equal to the side of the businesses and its customers. Web applications became famous because it does not require the user to install any additional software to access the service it is just one click away on their browser no matter what device (mobile, laptops or desktops) they are using. China yacht industry will maintain a market growth rate of no less than 30% per year, China is expected to seize the opportunity to welcome the future transfer of the yacht industry and become a new focus of the world's yacht industry[2].With the increasing numbers of the yacht, the needs for the existing yacht to repair became another likely service needed by the owners which later became the part of the YSC, with also the increasing numbers of repair to be done yacht charters start to employ repairmen to offer yacht repair and maintenance services.

On the other hand findings of IBIS World say that “The revenue of repairs and maintenance of the yacht is expected to rise over the years till the year 2021[3]”. The two studies both say that the market on yacht servicing is unsaturated; owners of yacht are ready to find charters that offer yacht repair and ready to cross the borders to avail the service. Hurley states numbers and statistics regarding the needs for the yacht servicing and systems that already exist in the market that is also used by different yacht charter [4] , which also corresponds to the research during the year 2018 have conducted in china that the need and demand for e-platforms for yacht related business are high[5].Progressive web apps are now on the demand and are being popularly used nowadays because of their similarities on mobile applications, WANG, ZHENG, and CHEN findings on the research entitled Analysis of the Capital Composition for Yacht Industry in China tell that the rapid development of the yacht industry bring different problems such as difficulty to adapt the changes happening on the current market, both of Wang and his fellow researcher in China presents the existing problems such as strengthening the market with the use of technology such as cloud and the web itself to cater a wide array of services[6].

Researchers from the California surveyed 130 people who had downloaded the RideScout app to evaluate attitudes and perceptions toward mobile apps, travel behaviour, and modal shift [8]. Mobile apps and web app became a trendy way to make it easy to bring marina services on the hand of the clients and service provider, although some issues has been found such as security and privacy. To cope up with the business, service management platforms are created to deal with untidy management of transactions and data. While on the other hand adding new features to make it easy for the future end user of the system are continued to discover. Mobile apps or web apps the key here is the accessibility, the ability of the systems itself to adjust to the environment of the platform it is being access. The technology and social networking revolution in America is transforming every area of American life, including transportation. The advent of a range of new technology-enabled transportation services that take advantage of mobile communications technology has been one of the most significant shifts. Americans will have new mobility options because to technological advancements and social networking technologies. New, technology-enabled transportation services might eliminate the need for many of these services by providing Americans with more mobility options and better methods to explore these options owning a personal car is becoming more common among Americans, resulting in a considerable drop in vehicle travel.

This research wants to imply the benefits of using a Service-oriented Information system on the Boat Repair and Rental that may deal with yacht charter, local Bangka owner and tourists with the use of developmental research strategy and instrumentation such as User acceptance tests, besides, that this research also wants to gather recommendations for future improvements of the research, another goal of this Research is to develop a service oriented software that can solve problems encountered in a Boat repair and rental in terms of scheduling and management of services such as status updates of repair in email and easy report generation .Boat reservation and integrating geolocation API and features.

4. MATERIALS AND METHODS

The researchers will use the descriptive - developmental method of research in this study. Descriptive research was used to obtain information concerning the current status of the phenomena to describe "what exists" with respect to variables or conditions in a situation, as these enable statistical analysis to be made while developmental because of the proposed Service-oriented system that is to be develop. After the development phase, the Service-Oriented system will be evaluated by selected respondents. This method of research applied prominently in this study because it presented the problems the company is facing in terms of management, scheduling and report generation.

To determine the effectiveness of the software, a non-randomized type of sampling is applied to increase the accuracy of results, the chosen respondents in the research are those who have knowledge in the yacht cluster servicing and those who are in the target market of the business specified.

In developing the said service-oriented platform the researchers will use the Agile model as the framework in developing and finishing the project. This is development method is popularly used in development a system because it covers the loop holes the waterfall has such as difficulty in incremental changes.

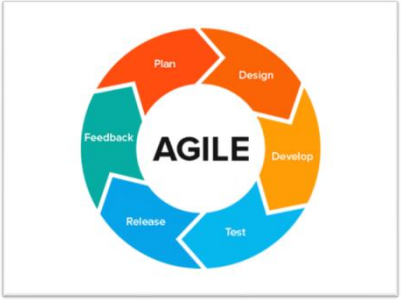

Figure 3.1: Agile Model

Figure above shows the Agile model. Agile Methodology is a people-focused, results-focused approach to software development that respects our rapidly changing world. It's centred on adaptive planning, self-organization, and short delivery times. It's flexible, fast, and aims for continuous improvements in quality [9].This method suits to the system that will be developed because this method can go back and forth to the previous and succeeding phases of the system development.

Plan

This phase is important because it will serve as the foundation of the development of the system and with this, the Researchers will do benchmarking and conduct a preliminary investigation on the existing business process of the stakeholder or client they are dealing with. In this phase also defines the scope of the system and system requirements.

Design

The process of defining the components, modules, interfaces, and data for a system to satisfy specified requirements. After the information has been gathered the Researchers will deal with the designing phase this phase includes the possible system architecture and the software tool needed to develop the said software.

Develop

This phase is more on the technical side of the process in the system, the Researchers will deal with programming and defining how to automate the business process in terms of translating it into a system requirement, in this phase also may include a prototype of the system and versions of it upon revision or reworking.

Test

After the development has been finished the output will undergo the testing process, different series of testing will be done to provide back job on the side of the developers. The Researchers will also deal on concluding what must be and what must not appear on the holistic features of the system.

Release

A release phase deals with some sort of experimental procedures even though the system passed the testing phase, it will still need to undergo on the experimentation because the environment of testing and operational is different, this phase also wants to know the feedback of the future end users of the system.

Feedback

In feedback this wants to know what to add, remove and remain on the features present on the system, the feedback also want to get the insights of the end users and deal with different issues such as data anomaly and Usability rating of the system, so that the output is ready to be deployed on the actual operational environment of the organization. after identifying the highest possible count of fish.

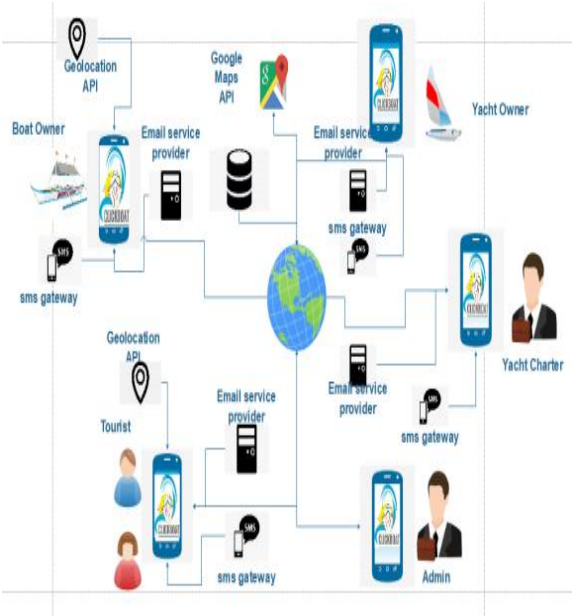

Figure 2. System architecture of the Developed Application

Table 1. Mobile App Requirements

Hardware	Technical Requirement	Purpose
Android	Android devices with Android 7+	Operating system
RAM	2gb above	Memory
ROM(Read Only Memory)	16 gb above ROM	Memory

5. RESULTS AND DISCUSSION

Problems encountered by the respondents on the existing system

Formula for getting the percentage:

Y = P% * X

Table 2. Tourist itinerary

Problem	Frequency	Rank
Finding a good boat driver for the boat ride service is difficult.	39	1st
Finding a good destination to go with is hard.	35	2nd
Finding a destination using Google search alone is difficult.	25	3rd
Searching for comments and feedbacks from other tourists is difficult.	17	4th
Others ; • price are not cascaded on boat rides	3	5th

As shown in the table above the major problem faced by the respondents in terms of tourist itinerary is "*Finding a good boat driver for the boat ride service*" which is answered by the 39 (97.5) of the respondents it can be assumed that it is an untidy job or there is no actual platform that deals with this type of business process same goes by the difficulty in "*Finding a good destination*" which is answered by 35 (87.5) of the respondents while the problem on "*Searching for comments and feedbacks from other tourists*" is answered by the 17 (42.5) respondents and using only Google maps as a platform gets 25 (62.5) answers from the respondents which can be assumed that the lack of platforms for tourist itinerary is a problem for tourists.

Table 3. Boat scheduling

Problem	Frequency	Rank
Deal with bulk transactions especially during peak seasons.	20	1st
Monitoring the good service provided by the boat owner to tourist is not recognized.	19	2nd
Managing boat reservation and scheduling transaction in manual manner.	17	3rd
Others	0	0

The respondents were asked what are the problems they have encountered in scheduling of the boat ride service for the tourist, the results as show on the table that 20 (100%) of the respondents have experienced "*difficulty in dealing with bulk transactions especially during peak season*"; while 19 (95%) of the respondents have find it difficult on "*Monitoring the good service provided by the boat owner to tourist*" and 17 (85%) of them have find it difficult in terms of "*Managing boat reservation and scheduling transaction in manual manner*".

Table 4. Yacht Repair Scheduling

Problem	Frequency	Rank
Yacht repair appointment in manual manner is difficult.	10	1st
Yacht repair appointment is challenging due to service fee are not cascaded.	10	1st
Some repair updates of yacht is not timely	8	2nd
Others	0	0

The respondents were asked to indicate what are their problems on yacht repair scheduling, 10 (100%) of the respondents answered "*difficulty in the manual process*"; 8 (80%) of the respondents have answered "*repair status update is not timely*" and also 10 (100%) have said that the "*repair scheduling is*

challenging due to the reason of service fee are not cascaded".

Table 5 . Yacht Repair management

Problems encountered	Rank
Using Log books and Record books to create a report is difficult.	1st
Difficulty to create sales invoice in a manual manner.	2nd
It takes so much time in preparing reports like annual reports, sales reports and invoice lists of finished transaction.	3rd
Difficulty in syncing information from different sources	4th

The table above shows the problems that the yacht charter have experienced using the manual system, Renato F. Lisud the manager of the Papaya Yacht Charter and Services Inc. said that although there is a system they are using in the company they find it hard to use it because it needed expertise in terms of accounting and proficiency in using flat database management systems such as Microsoft excel, one of the hindrance he also take note "the business lacks automation different clients are wanting to have the possible and easiest way to access the yacht repair services". As the researcher have demonstrate the technology solution to the Charter manager he have seen how the notification features of the system work on both Email and SMS he wants to have those kind of automation around their business saying that he wanted to implement it on the organization.

Modules of the System developed to address the common problems encountered by the respondents

Table 6. Modules of System developed to address the problem

Module name	**Access and Capability**
Tourist	• List of Destinations with current weather status, ratings and feedbacks. • List of Boat owner for boat reservation selection. • Geolocation View of Destination and current User location. • Providing rating for destinations and boat drivers • Crud functionalities on boat reservation such as Create, Cancel and update. • Receive Email and SMS notifications about transaction status. • Add proof of Down payment or Reservation fee for transaction assurance. • Password changing and retrieval. • User's profile
Boatdriver	• List of Boat reservation requests. • List of Previous Transactions and Tourist's rating. • Map view of the destination the tourist wanted. • Capability to update the Boat reservation's status. • Received SMS and Email notification about the status of boat reservation.
Boatdriver	• Verify the reservation fee or down payment of

	the tourist. • Password updating and retrieval facility. • View my profile and see data visualization of transactions
Yacht Owner	• Create, update and cancel a repair appointment. • View repair invoices. • Add repair feedback. • Receive invoice copy of repair in Email and SMS. • My profile • Password Retrieval and Updating.
Yacht charter	• List of repair appointments • Update Repair Status and generate invoices in PDF attached in the email notification to the clients. • Generate reports and view simple data visualization. • Calendar views of repairs. • View client's feedbacks. • My profile • Updating and Retrieval of Password. • Manage repairs and Services.

The table 8 shows the modules of the system the Researchers have developed to address the problems of the respondents shown in table 4, 5, 6 and 7 which describes the problem they encountered in using the manual or traditional business process. With these the Researchers have come up to the idea of creating modules that will automates the business process in terms of report generation and transaction awareness through system generated notifications that involved SMS and Email.

Tools and technology used to develop the solution to the problem

Table 7. Tools and technology used to develop the system

Tool or Technology used	**Type**	**Purpose**
Text editor tool	computer program	Edit plain text file, scripts and batch commands.
Xammp	cross-platform web server solution	Consisting mainly of the Apache HTTP Server, MariaDB database, and interpreters for scripts written in the PHP and Perl programming languages.
PHPmailer	Php class library file	PHPMailer is a code library to send emails safely and easily via PHP code from a web server.
DomPDF	Php class library file	HTML to PDF converter tool.
Bootstrap	CSS(Cascading Style Sheet) Framework	CSS Framework for developing responsive and mobile-first websites.
Filezilla	FTP client Software	Deploy Web app source codes such as Php,CSS and Javascript files on the Web Server.
Cpanel	Web-hosting control Panel	Manage Web apps configurations

		such as SSL certificates and other managerial task on website management,
Swing2app	Mobile App Builder	Create or Convert Web application into native mobile application.
Itextmo	SMS Gateway API	Send SMS notification in a specific mobile numbers.
OpenWeather	Weather API	Fetch weather updates and statuses.
Google Maps API	API	Fetch web mapping and geographic view of the specific location.
OpenStreetmaps	API	Create a free editable geographic database of the world.

The table above shows the technology and tools the Researchers used to develop the software in addressing the problem faced by the respondents, majority of these tools are used in creating progressive web apps and APIs are only used to give additional features on the system such as notification like Phpmailer and Itexmo does, while there are also present Geographic mapping tools used in the development such as Openstreetmaps and Google maps API.

Level of acceptance of the respondents in the developed system "Click Boat: Boat Repair and Rental Management System" in terms of?

5r=Number of Responses with 5

4r=Number of Responses with 4

3r=Number of Responses with 3

2r=Number of Responses with 2

1r=Number of Responses with 1

T=Total number of Respondents

Formula:

Mean= ((5r*5)+(4r*r)+(3r*3)+(2r*2)+(1r*1))/T

Mean2=((5r*25)+(4r*16)+(3r*9)+(2r*4)+(1r*1))/T

Standard Deviation=√ (Mean2-Mean)

Average= (∑mean)/number of parameters

Table 8. System Acceptability Rating In Terms Of User Friendliness

Parameters	Mean	Descriptive Equivalent
The system can adapt on different screen sizes.	4.68	Highly Acceptable
The UI design is learnable and easy to use/operate	4.58	Highly Acceptable
The system is easy to navigate and operate.	4.70	Highly Acceptable
The system provides different interactive views of information.	4.66	Highly Acceptable
The system provides a convenient way of password retrieval process.	4.58	Highly Acceptable
The system interface is aesthetically designed	4.66	Highly Acceptable

The table above shows the level of acceptance of the respondents in terms of the system's user-friendliness it shows a Standard deviation that is not below or greater than 4 which means that the data is not well dispersed or spread out. The mode is either 5 or 4 which tells moderate and high acceptability of the system's User friendliness. The traditional tourism industry is unable to follow with the development trend and with the emergence of modern personalized information services, and the service provided by the tourism e-commerce environment also important for improving the tourists' satisfaction and also the competitiveness [10].A different system has been on the trend to provide newer experience to the target market to avoid churns and decline of possible clients.

Table 9. System Acceptability Rating In Terms Of Functionality

Parameters	Mean	Descriptive Equivalent
The system provides accurate ratings, insights and data visualization	4.76	Highly Acceptable

The system improves the way the business process works.	4.63	Highly Acceptable
The system is capable of handling errors on some cases like invalid inputs such as dates and text strings.	4.63	Highly Acceptable
The system can aid the user to finish a specified task.	4.60	Highly Acceptable
The system can facilitate the accomplishment of specified tasks and functions.	4.62	Highly Acceptable

The table above shows the level of acceptance of the respondents in terms of the system's Functionality it shows a Standard deviation that is not below or greater than 4 which means that the data is not well dispersed or spread out. The mode is either 5 or 4 which tells moderate and high acceptability of the system's Functionality. While the incorporation of some earlier RAISA technologies such as the ATM may have radically changed the way the customer experiences [11].The concept of this information system is to automate the system even though the automation itself is not in the form of holistic automation but automating some of the most vital parts of the process such as analytics, notifications and other informatics related function that this system can bring on user and persuade to use newer systems.

Issues encountered during the deployment of the System

Table 10. Deployment Issues

Issue	Cause	Severity
DNS error	Web server Downtime	Major
MySQL error	Web server Downtime or TCP/IP socket not listening	Moderate
Web Mail Not working	email account has exceeded its quota IMAP/POP is not enabled.	Minor
Website suspension due to usage limits	Exceeded web server resources.	Minor
FTP service not working	Webserver Downtime	High
GPS not working	GPS service is not working on non-encrypted connection like HTTP (port 80)	Cosmetic
App not installed	Android version is not compatible to the app package.	Moderate
Cron Jobs Not working properly	Execution time exceeded. Incorrect permission.	Moderate
Website not safe alert	SSL certificate problem, DNS propagation in progress	Moderate

The following data on the table shows different issues that the Researcher have encountered during the deployment of the System that is according to the ISQTB (International Software Testing Qualifications Board), to sum it up, it can be seen can seen that a web based systems may encounter unexpected problems such as downtime due to multiple services and requests it is receiving across the network. In the 24/7 web industry, business owners can feel a constant need to push the latest and greatest software to the live site in order to remain competitive and ensure growth (John ,Mostashari and Mansouri,2011). It is important to always have a backup and select the best Web service package service as soon as possible because cheap services may bring drawbacks as the system scales up such as increasing volume of users and data that it process, In addition to that cross platform system such as combination of Native and mobile apps are a bit complicated due to the way they access the data and present it to the users.

System features recommended by the respondents

Table 13. Recommended features

Features	Frequency	Rank
Automated SMS and Email notification	67	1st
Payment Gateway Facility	62	2nd
Google Maps Direction	57	3rd
Simple analytics	39	4th

The implication of this finding is the respondents have suggests different features

that may improve the system for later purposes, to explain that the respondents wanted to automate almost the entire process on the business which involved notification features. The payment facility is the second majority feature they want to be added, with these result we can assume that the need of the users for automation for payment process is a must implemented feature on the system, On the Geographic features the respondents wanted to add the Google map direction services because this gives a better routes on the users while the analytics have the least number of recommendation it might be assumed that respondents have less experience or knowledge to the analytics itself.

6. CONCLUSIONS

From the above summary, the following can be concluded from this study:

- A manual process on boat repair and rental is difficult to deal especially as the data or information that the user is managing scales up especially on Boat rental where the transaction needed to process increases on peak seasons, tourists have a problem where destination to go and who else to contact to access the service. While on the other hand on yacht servicing clients that needed to avail repair services face challenges in creating a repair appointment due to price not being cascaded, paper-dependent processes, and lastly the repair management of Yacht charter is difficult because of different data sets from different flat file or paper sources that may involve logbooks, stand-alone system, and flat databases they are using, the need for automation can be seen on the problems they are facing that might resolved by the creation of a platform that can facilitate report generation and notification automation.
- The need for the development of a service-oriented platform that is intended to be used by tourists, yacht owners, boat drivers, and yacht charters where they can avail and provide services across the internet to adopt the new generation of age of paperless businesses.
- The internet itself plays a big role in the SDLC process it provides a readymade library for a fast phased development process that involves frameworks, API, libraries, and other existing systems that developers may benchmark to create innovative solutions for businesses and the community. Software severities are expected to arise on initial deployment due to incompatibility of legacy and newer systems
- Based on the findings of the acceptance test done, the study shows that it is difficult to measure the user-friendliness of the system due to the issues that users are not or does not have the same standards on Graphical interface and Cognitive understanding. The correct selection of colors, fonts and symbols must be considered to provide a user-friendly system. On the other hand, they find out that users wanted to have access to details about the information itself that may include rating and also they seek more on automation modules that may give them ease on accessing the service and the information.
- Good planning especially from a phase to phase basis such as holistic planning of deployment and procurement of resources must also discuss carefully. A series of tests from the different environments must be done first to assure the quality and assurance aspect of the system. The discrepancy between how the actual business environment and the unit testing environment is large. Software severity is a helpful tool to measure the defects and bugs of the software. In addition to that the because the deployment is one of the most vital processes of SDLC choosing the best package on technology to integrate is the best decision to do to avoid back jobs and troubles such as lack of bandwidth or performance of the system.
- The software evaluation of the study discovers that users wanted to have everything in touch of their hand; they wanted everything to be processed on just simple steps and clicks. Geographical and automated features are those features they want to have for the system improvement. There was evidence on the results that the system has a high acceptability rating on the respondents and room for its improvement is still possible especially if there is a budget for implementation and technology procurement.

7. RECOMMENDATIONS

The following are recommendations for future researches and implementation:

1.The developed system may improve its usage on a larger web hosting packaged.

2.Project budget is a very vital part of a study with these we can expand the quality of the system that we are going to develop.

3.The developers may add search box and sort the transactions based on its status and date creation.

4.Future developers may include features that can reach out other related people even though they are not the user of the system.

5.The future Researcher may use this paper as a reference related to business process automation related to Marina businesses. Accessing the system.

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